

Incompressible flow as redistribution of an accumulated field: behavior across closures, exact balances, and a self-calibrating prediction

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Abstract

This manuscript studies a single incompressible continuum model and characterizes its behavior across its main parameter closures. The model carries two vector fields: an accumulated field $\mathbf{D}(\mathbf{x}, t)$ and its rate $\mathbf{P}(\mathbf{x}, t) = \partial_t \mathbf{D}$, closed by a pressure Π enforcing $\nabla \cdot \mathbf{P} = 0$. The accumulated field is allowed to have nonzero initial content, $\mathbf{D}(\mathbf{x}, t) = \mathbf{D}_0(\mathbf{x}) + \int_0^t \mathbf{P}(\mathbf{x}, s) ds$, so stored energy can be present at $t = 0$. Relative to the incompressible Navier–Stokes structure the system adds two independent mechanisms acting on different objects: a restoring term $c^2 \Delta \mathbf{D}$ on the accumulated field and a linear friction $-\lambda \mathbf{P}$ on its rate. The restoring term represents an elastic storage channel; the friction term represents a bulk relaxation channel and may occur independently of elastic storage. Setting $c = \lambda = 0$ returns incompressible Navier–Stokes exactly, with $\mathbf{P}$ the velocity. Setting $c = 0, \lambda > 0$ returns Navier–Stokes with linear drag, not a storage medium. The remaining storage closures include a conservative wave, one-cutoff damped regimes, and the full inertial–dissipative regime. Three exact balances govern the dynamics: an energy identity in which the restoring term is conservative rather than dissipative, an enstrophy identity with a conservative orientation store, and an exchange bound capping how fast the stored and active orientation channels trade. The model’s sharp linear prediction is a two-cutoff finite band of wavenumbers that relax by oscillation when $\nu > 0, \lambda > 0$, and $c^2 > \nu \lambda$; the one-channel limits produce only one cutoff. The band edges plus one in-band frequency invert to all three parameters, so the prediction is self-calibrating in the linear regime and over-determined by several independent measurements. A scalar numerical study verifies the internal linear balances, the diffusion limit, the band, and the inversion formulas. The one hypothesis held provisional — the small-deformation rate law — is then crossed: the naive objective upgrade is shown to break the energy budget irreparably, while a conformation-tensor formulation closes it exactly, at the stated cost that the accumulated field becomes the linear shadow of a strain-state tensor. All results are stated and proved without reference to any external framework; a closing section offers one optional interpretation, fully detachable, that no result depends on.

Keywords: incompressible flow; Navier–Stokes; memory variables; viscoelasticity; modal ringing; energy and enstrophy identities.

Contents

1	What the model is	2
2	The system	3

3 Behavior across closures	4
4 The Newtonian closure: exact containment	5
5 Energy identity: the restoring term is conservative	5
6 Enstrophy identity and the orientation store	5
7 Orientation-transfer bound	6
8 The finite band: a self-calibrating prediction	6
8.1 Parameter inversion	7
8.2 Over-determination	7
9 Modal phase: stored versus active	8
10 Vorticity and the three-dimensional obstruction	8
11 Pressure as absorption of the longitudinal current	8
12 Numerical study	8
12.1 Two-dimensional incompressible vector validation	9
13 Finite-deformation extension: the boundary of the small-deformation regime	10
13.1 The naive convected law is impossible	11
13.2 A conformation tensor closes the budget	11
13.3 Thermodynamic consistency with all dissipative channels	12
13.4 What the extension reveals, and at what cost	13
14 Falsifiers	13
15 Gap register	15
16 An optional interpretation (detachable)	15
A Energy identity	16

1 What the model is

We work on a domain $\Omega \subset \mathbb{R}^d$, $d \in \{2, 3\}$, either a periodic box or a smooth bounded domain with boundary conditions chosen explicitly so that the integrations by parts below carry no boundary terms. A sufficient bounded-domain choice is homogeneous no-slip rate $\mathbf{P}|_{\partial\Omega} = 0$ together with a compatible accumulated field condition such as $\partial_n \mathbf{D} = 0$ for the restoring store; the periodic case is the default setting for all modal statements. More general boundary conditions are admissible only if they cancel the same boundary fluxes in the energy and enstrophy identities.

The model is built from two time-dependent vector fields and a scalar. The first field, $\mathbf{D}(\mathbf{x}, t)$, is the *accumulated field*: at each point it records an initial stored state plus the running integral

of the local rate of flow,

$$\mathbf{D}(\mathbf{x}, t) = \mathbf{D}_0(\mathbf{x}) + \int_0^t \mathbf{P}(\mathbf{x}, s) ds, \quad \partial_t \mathbf{D} = \mathbf{P}.$$

Allowing $\mathbf{D}_0 \neq 0$ is essential: it permits stored energy at $t = 0$ even when the active rate initially vanishes. The second field, $\mathbf{P}(\mathbf{x}, t) = \partial_t \mathbf{D}(\mathbf{x}, t)$, is the *rate field* or *active field*; it plays the role the velocity plays in ordinary fluid mechanics. The scalar $\Pi(\mathbf{x}, t)$ is a pressure: it is whatever field is required, at each instant, to keep $\mathbf{P}$ divergence-free. We call a vector field *transverse* when it is divergence-free, $\nabla \cdot \mathbf{P} = 0$; mode by mode this reads $\mathbf{k} \cdot \mathbf{P}_{\mathbf{k}} = 0$. Unless otherwise stated we also take the initial accumulated field to be transverse, $\nabla \cdot \mathbf{D}_0 = 0$, so $\nabla \cdot \mathbf{D} = 0$ for all time. This is a simplifying sector choice, not the claim that a longitudinal preload stores no energy. If $\mathbf{D}_0$ has a Helmholtz-longitudinal part, it can contribute to the elastic energy $\frac{1}{2}c^2 \|\nabla \mathbf{D}\|^2$, but it has zero curl and therefore contributes nothing to the orientation/enstrophy store $G = \frac{1}{2}c^2 \|\nabla \Omega_{\mathbf{D}}\|^2$. In the incompressible $\mathbf{P}$ -dynamics, the restoring force generated by this longitudinal part is a gradient and is absorbed by the pressure; it is therefore a decoupled elastic-energy channel outside the transverse modal sector studied here. We impose $\nabla \cdot \mathbf{D}_0 = 0$ so the storage analyzed below lives in the same transverse sector as the incompressible dynamics and the vorticity identity.

These are the only primitives. “Primitive” here means simply a field posited at the outset, from which everything else is computed; no field below is independent of $\mathbf{D}$, $\mathbf{P}$, Π . The phrase *closure* is used in two standard senses, both made explicit where they occur: the incompressibility closure (the constraint $\nabla \cdot \mathbf{P} = 0$ together with the pressure that enforces it), and a *parameter closure* or *regime* (a fixed choice of the coefficients ν, c, λ below, within which the dynamics is autonomous). The behavior of the model “across closures” means its behavior across these regimes, and is the subject of the paper.

Nothing in this section, or in any result, requires deciding what $\mathbf{D}$ “ultimately is.” The mathematics is fixed once the fields and the equations are fixed. A single optional reading of $\mathbf{D}$ is offered in Section 16, after every result is established, and is explicitly detachable.

2 The system

The fixed-regime system is

$$\begin{aligned} \partial_t \mathbf{D} &= \mathbf{P}, \\ \partial_t \mathbf{P} + (\mathbf{P} \cdot \nabla) \mathbf{P} &= -\nabla \Pi + \nu \Delta \mathbf{P} + c^2 \Delta \mathbf{D} - \lambda \mathbf{P} + \mathbf{S}, \\ \nabla \cdot \mathbf{P} &= 0, \end{aligned} \tag{1}$$

with three nonnegative parameters — a gradient (solvent-type) viscosity $\nu \geq 0$, a restoring stiffness $c \geq 0$ with the dimensions of a wave speed, and a bulk friction rate $\lambda \geq 0$ — and a forcing $\mathbf{S}$. The role of each term is fixed by which object it acts on:

Term	Acts on, and how
$\partial_t \mathbf{D} = \mathbf{P}$	defines the active field as the rate of the accumulated field
$(\mathbf{P} \cdot \nabla) \mathbf{P}$	advection: the active field transports itself
$-\nabla \Pi$	incompressibility closure
$\nu \Delta \mathbf{P}$	gradient dissipation of the <i>active</i> field
$c^2 \Delta \mathbf{D}$	restoration of the <i>accumulated</i> field (shown conservative, §5)
$-\lambda \mathbf{P}$	bulk friction on the <i>active</i> field
$\mathbf{S}$	external forcing

The single structural choice that generates everything below is that the restoring force acts on the *accumulated* field $\mathbf{D}$, while *both* dissipative forces act on its *rate* $\mathbf{P}$. The two dissipative terms scale differently with wavenumber — one constant, one growing as νk^2 — and that mismatch is the origin of the central prediction.

Scope of the rate law. The law $\partial_t \mathbf{D} = \mathbf{P}$ is a local, fixed-frame, small-to-moderate-deformation closure. It is the law under which the exact energy identity of Section 5 holds, and the regime in which every result below is stated. A finite-deformation transport law for $\mathbf{D}$ (convected or corotational) modifies the energy bookkeeping; Section 13 resolves that boundary of the present regime, showing which finite-deformation upgrade is impossible and which one preserves an exact energy law, at a stated change of primitive. Every result in Sections 3–12 is stated and proved within the small-deformation regime and is unaffected by that extension.

3 Behavior across closures

Each choice of (ν, c, λ) is a regime with a definite character. The table below is the backbone of the paper; each row is established in the section cited, and the linear character of each corner is exact (advection and pressure suppressed, valid mode by mode for transverse shear modes). The table separates two questions that were easy to conflate: whether the medium has a storage channel ($c > 0$) and whether the active field has bulk drag ($\lambda > 0$).

Closure	Reduced equation for $\mathbf{D}$ or $\mathbf{P}$	Character
$c = \lambda = 0$	$\partial_t \mathbf{P} + (\mathbf{P} \cdot \nabla) \mathbf{P} = -\nabla \Pi + \nu \Delta \mathbf{P}$	exact incompressible Navier–Stokes (§4)
$c = 0, \lambda > 0$	$\partial_t \mathbf{P} + (\mathbf{P} \cdot \nabla) \mathbf{P} = -\nabla \Pi + \nu \Delta \mathbf{P} - \lambda \mathbf{P}$	Navier–Stokes with linear drag/Brinkman-type relaxation; no elastic storage
$c > 0, \nu = \lambda = 0$	$\partial_{tt} \mathbf{D} = c^2 \Delta \mathbf{D}$	conservative accumulated-field wave (§12)
$c > 0, \nu = 0, \lambda > 0$	$\partial_{tt} \mathbf{D} + \lambda \partial_t \mathbf{D} = c^2 \Delta \mathbf{D}$	wave with bulk drag; lower-cutoff ringing only (§12, §8)
$c > 0, \lambda = 0, \nu > 0$	$\partial_{tt} \mathbf{D} - \nu \Delta \partial_t \mathbf{D} = c^2 \Delta \mathbf{D}$	rate-gradient damping; upper-cutoff ringing only (§12, §8)
$c > 0, \nu > 0, \lambda > 0$	$\partial_{tt} \mathbf{D} - \nu \Delta \partial_t \mathbf{D} + \lambda \partial_t \mathbf{D} = c^2 \Delta \mathbf{D}$	full inertial–dissipative storage regime; finite two-cutoff band iff $c^2 > \nu \lambda$ (§8)

No storage corner reduces to a textbook one-dimensional rheological element, because the restoring force acts on $\mathbf{D}$ while the dissipative forces act on $\partial_t \mathbf{D}$; the names “wave,” “drag,” and “rate-gradient” are descriptive, not identifications. The case $c = 0, \lambda > 0$ is included to avoid overclaiming: it is a legitimate drag closure, but it is not a storage-and-return medium. The rest of the paper establishes the balances that hold across the stated regimes.

4 The Newtonian closure: exact containment

Set $c = \lambda = 0$. Then $\mathbf{D}$ decouples from the $\mathbf{P}$ -equation, which becomes

$$\partial_t \mathbf{P} + (\mathbf{P} \cdot \nabla) \mathbf{P} = -\nabla \Pi + \nu \Delta \mathbf{P} + \mathbf{S}, \quad \nabla \cdot \mathbf{P} = 0 :$$

incompressible Navier–Stokes, with $\mathbf{P}$ identified as the velocity. This is a property of the construction, not an independent finding: the two added terms were built to vanish in this corner. We record it as the consistency requirement that anchors the parameter space — the model is an extension of incompressible Navier–Stokes, exactly Newtonian when the storage and friction coefficients are zero, and departing from it only when they are not.

5 Energy identity: the restoring term is conservative

Define the energy

$$E(t) = \frac{1}{2} \int_{\Omega} (|\mathbf{P}|^2 + c^2 |\nabla \mathbf{D}|^2) \, d\mathbf{x}.$$

Result 1 (Energy balance). *For smooth solutions of (1) under periodic boundary conditions, or under bounded-domain conditions that cancel the same boundary terms (for example $\mathbf{P}|_{\partial\Omega} = 0$ with compatible accumulated-field boundary data),*

$$\frac{dE}{dt} = -\nu \int_{\Omega} |\nabla \mathbf{P}|^2 \, d\mathbf{x} - \lambda \int_{\Omega} |\mathbf{P}|^2 \, d\mathbf{x} + \int_{\Omega} \mathbf{P} \cdot \mathbf{S} \, d\mathbf{x}.$$

The derivation (Appendix A) is the standard L^2 pairing: the pressure term vanishes by incompressibility, the advective term by the usual cancellation, and the restoring term cancels exactly against the time derivative of $\frac{1}{2}c^2 \|\nabla \mathbf{D}\|^2$. Three consequences:

The restoring channel stores, it does not dissipate. The term $c^2 \Delta \mathbf{D}$ contributes nothing to the right-hand side; it moves energy into and out of the gradient store $\frac{1}{2}c^2 \|\nabla \mathbf{D}\|^2$ and returns all of it. It is therefore not a regularizer. In particular the model does not act on $\partial_t \mathbf{P}$ the way a rate-regularization would, and inherits no well-posedness from such; the added elastic term acts on the accumulated history $\mathbf{D} = \int_0^t \mathbf{P}$ and is energetically neutral, term by term in the balance.

The two dissipative channels are non-redundant. They remove energy at rates $\nu \|\nabla \mathbf{P}\|^2$ (small scales) and $\lambda \|\mathbf{P}\|^2$ (the bulk, including the gradient-free mode). Their distinct wavenumber scaling, νk^2 versus λ , produces the finite band of Section 8.

It is a bookkeeping law. Every unit of energy is active, stored, dissipated, or forced. Nothing appears or leaks.

6 Enstrophy identity and the orientation store

Let $\boldsymbol{\omega} = \nabla \times \mathbf{P}$ be the active vorticity and $\boldsymbol{\Omega}_{\mathbf{D}} = \nabla \times \mathbf{D}$ the *accumulated* (stored) vorticity, so that $\partial_t \boldsymbol{\Omega}_{\mathbf{D}} = \boldsymbol{\omega}$. Taking the curl of (1),

$$\partial_t \boldsymbol{\omega} + (\mathbf{P} \cdot \nabla) \boldsymbol{\omega} = (\boldsymbol{\omega} \cdot \nabla) \mathbf{P} + \nu \Delta \boldsymbol{\omega} + c^2 \Delta \boldsymbol{\Omega}_{\mathbf{D}} - \lambda \boldsymbol{\omega} + \nabla \times \mathbf{S}.$$

Result 2 (Enstrophy balance). *With $W = \frac{1}{2} \|\boldsymbol{\omega}\|^2$ and $G = \frac{1}{2} c^2 \|\nabla \boldsymbol{\Omega}_{\mathbf{D}}\|^2$,*

$$\frac{d}{dt} (W + G) = \int_{\Omega} \boldsymbol{\omega} \cdot (\boldsymbol{\omega} \cdot \nabla) \mathbf{P} - \nu \|\nabla \boldsymbol{\omega}\|^2 - \lambda \|\boldsymbol{\omega}\|^2 + \int_{\Omega} \boldsymbol{\omega} \cdot (\nabla \times \mathbf{S}).$$

In two dimensions the stretching term $\int \boldsymbol{\omega}(\boldsymbol{\omega} \cdot \nabla) \mathbf{P}$ vanishes identically, and unforced $\frac{d}{dt} (W + G) = -\nu \|\nabla \boldsymbol{\omega}\|^2 - \lambda \|\boldsymbol{\omega}\|^2 \leq 0$.

The derivation mirrors Appendix A: the advection term drops by incompressibility and $c^2 \int \boldsymbol{\omega} \cdot \Delta \boldsymbol{\Omega}_{\mathbf{D}} = -c^2 \partial_t (\frac{1}{2} \|\nabla \boldsymbol{\Omega}_{\mathbf{D}}\|^2) = -\dot{G}$. The term $G = \frac{1}{2} c^2 \|\nabla \boldsymbol{\Omega}_{\mathbf{D}}\|^2$ is a *conservative store of accumulated vorticity*, exactly parallel to the gradient store in Result 1. In three dimensions the only source of growth is the stretching term $(\boldsymbol{\omega} \cdot \nabla) \mathbf{P}$ — the same term that carries the classical three-dimensional difficulty (Section 10); no regularity is claimed.

7 Orientation-transfer bound

The two channels W (active) and G (stored) are not separately conserved; they exchange. Differentiating G and using $\partial_t \boldsymbol{\Omega}_{\mathbf{D}} = \boldsymbol{\omega}$,

$$\dot{G} = c^2 \int_{\Omega} \nabla \boldsymbol{\Omega}_{\mathbf{D}} : \nabla \boldsymbol{\omega} =: X(t),$$

the same quantity entering $\dot{W}$ with opposite sign via $c^2 \int \boldsymbol{\omega} \cdot \Delta \boldsymbol{\Omega}_{\mathbf{D}} = -X$. Thus X is the only term coupling the channels; $X > 0$ loads the store, $X < 0$ feeds the active vorticity.

Proposition 1 (Transfer bound). *For smooth solutions under the stated boundary conditions,*

$$|X(t)| \leq c \|\nabla \boldsymbol{\omega}\| \sqrt{2G(t)},$$

with equality iff $\nabla \boldsymbol{\omega}$ and $\nabla \boldsymbol{\Omega}_{\mathbf{D}}$ are pointwise parallel.

Proof. Cauchy–Schwarz: $|X| = c^2 |\int \nabla \boldsymbol{\omega} : \nabla \boldsymbol{\Omega}_{\mathbf{D}}| \leq c^2 \|\nabla \boldsymbol{\omega}\| \|\nabla \boldsymbol{\Omega}_{\mathbf{D}}\| = c \|\nabla \boldsymbol{\omega}\| \sqrt{2G}$, since $\|\nabla \boldsymbol{\Omega}_{\mathbf{D}}\| = \sqrt{2G}/c$. $\square$

The cap is set by three transparent factors: the wave speed c ; the structure of the active vorticity $\|\nabla \boldsymbol{\omega}\|$; and the square root of the stored amount $\sqrt{G}$. The last gates the others: *if the store is empty, $G = 0$, then $X = 0$* — nothing can be released until something is accumulated. The first active factor gates preloaded orientation in the opposite direction: if $G(0) > 0$ but $\boldsymbol{\omega}(0) = 0$ and hence $\|\nabla \boldsymbol{\omega}(0)\| = 0$, then $X(0) = 0$. A preloaded orientation store cannot discharge as an instantaneous jump; release begins only as active vorticity acquires spatial structure. This is a Cauchy–Schwarz estimate on the coupling term, re-derivable in a line; its value is operational, as a falsifier (Section 14) and as the quantitative statement that the model transfers stored vorticity rather than creating it.

8 The finite band: a self-calibrating prediction

Linearizing (1) about rest, a single transverse Fourier mode of wavenumber k obeys

$$\ddot{D}_k + \gamma_k \dot{D}_k + \varpi_k^2 D_k = 0, \quad \gamma_k = \lambda + \nu k^2, \quad \varpi_k = c k. \quad (2)$$

The roots of $s^2 + \gamma_k s + \varpi_k^2 = 0$ separate oscillatory relaxation (ringing, $\varpi_k > \gamma_k/2$) from monotone relaxation ($\varpi_k < \gamma_k/2$). Writing $2ck > \lambda + \nu k^2$ as a quadratic in k :

Result 3 (Ringing regimes). *Assume $c > 0$. For the full two-dissipation case $\nu > 0$, $\lambda > 0$, a finite two-cutoff oscillatory band exists iff $c^2 > \nu \lambda$, and then exactly for $k \in (k_-, k_+)$ with*

$$k_{\pm} = \frac{c \pm \sqrt{c^2 - \nu \lambda}}{\nu}.$$

Inside the band the mode rings at $\Omega_k = \sqrt{\varpi_k^2 - \gamma_k^2/4}$ under the envelope $e^{-\gamma_k t/2}$. In the one-dissipation limits the oscillatory set is not a finite two-cutoff band: for $\nu = 0$, $\lambda > 0$ it is $k > \lambda/(2c)$; for $\lambda = 0$, $\nu > 0$ it is $0 < k < 2c/\nu$.

In the two-dissipation storage case, the finite band is a *channel counter*. With no restoring channel ($c = 0$) it is empty: no ringing at any k . With a restoring channel and a single dissipative channel constant in k ($\nu = 0$), the oscillatory set is a half-line with a lower cutoff only. With a restoring channel and a single channel growing as k^2 ($\lambda = 0$), it has an upper cutoff only. Only the two-channel structure — one rate constant, one rate $\propto \nu k^2$ — bounds the band on *both* sides, because at high k the gradient channel overwhelms the restoring frequency and relaxation returns to monotone. The presence, absence, and shape of the band read off the dissipative channel structure directly from linear response.

8.1 Parameter inversion

By Vieta's formulas applied to $\nu k^2 - 2ck + \lambda = 0$,

$$k_- + k_+ = \frac{2c}{\nu}, \quad k_- k_+ = \frac{\lambda}{\nu}. \quad (3)$$

Both edges fix the ratios c/ν and λ/ν . One absolute measurement — the ringing frequency Ω_{k^*} at any single in-band k^* — fixes the scale, giving all three parameters:

$$\nu = \frac{2\Omega_{k^*}}{\sqrt{(k_- + k_+)^2 k^{*2} - (k_- k_+ + k^{*2})^2}}, \quad c = \frac{1}{2}\nu(k_- + k_+), \quad \lambda = \nu k_- k_+.$$

The prediction is therefore self-calibrating: a resolved band produces the parameters with no free fitting, and the same parameters must then reproduce Ω_k and the envelope $e^{-\gamma_k t/2}$ at every other in-band wavenumber. The dimensionless control number is $\Lambda = c^2/(\nu\lambda)$, with ringing iff $\Lambda > 1$.

8.2 Over-determination

Once (ν, c, λ) are fixed, further linear features are forced with no freedom, each an independent measurement that must return the same three numbers. These are consequences a reviewer can re-derive in minutes; their worth is that each is a separate consistency test, not that any is a separate discovery.

Measured feature	Predicted value	Returns
Band edges $k_{\pm}$	$\frac{2c}{\nu}, \frac{\lambda}{\nu}$ via (3)	$c/\nu, \lambda/\nu$
Envelope decay rate vs k^2 (in-band)	affine: slope $\nu/2$, intercept $\lambda/2$	ν, λ
Sharpest-ringing wavenumber	$\sqrt{\lambda/\nu} = \sqrt{k_- k_+}$	λ/ν
Peak quality factor there	$Q_{\max} = \sqrt{\Lambda}/2$	Λ
Band ratio k_+/k_-	function of Λ only	Λ
Frequency-peak wavenumber	$\sqrt{2c^2 - \nu\lambda}/\nu$	combination

The envelope decay rate is affine in k^2 *only inside the band*, where every mode is underdamped and decays at $\gamma_k/2$; outside the band the relaxation is monotone with a different rate, so the affine fit must be restricted to $k \in (k_-, k_+)$. Mutual consistency of all rows is the content of the prediction; no row is a separate triumph.

9 Modal phase: stored versus active

For a single mode set $X_k = \varpi_k D_k$ (stored component) and $Y_k = \dot{D}_k$ (active component), so $E_k = \frac{1}{2}(X_k^2 + Y_k^2) = \frac{1}{2}R_k^2$ with $X_k = R_k \cos \theta_k$, $Y_k = R_k \sin \theta_k$. The unforced dynamics (2) is, exactly,

$$\dot{R}_k = -\gamma_k R_k \sin^2 \theta_k, \quad \dot{\theta}_k = -\varpi_k - \frac{\gamma_k}{2} \sin 2\theta_k, \quad \frac{dE_k}{dt} = -2\gamma_k E_k \sin^2 \theta_k \leq 0.$$

The decay law is the sharpest modal statement of the mechanism: *only the active component dissipates*. Fully stored ($\theta_k = 0$) the mode loses nothing; fully active ($\theta_k = \pi/2$) it dissipates maximally. A stationary angle exists iff $\varpi_k/\gamma_k \leq \frac{1}{2}$ (the monotone corner: the mode locks to a fixed storage/activity ratio); for $\varpi_k/\gamma_k > \frac{1}{2}$ the angle rotates and the mode perpetually exchanges stored and active components — which is the ringing of Section 8, seen from the phase side.

10 Vorticity and the three-dimensional obstruction

From the vorticity equation of Section 6, the accumulated field carries its own vorticity $\Omega_{\mathbf{D}}$, restored conservatively by $c^2 \Delta \Omega_{\mathbf{D}}$ and fed by the active vorticity; the friction $-\lambda \omega$ damps active vorticity uniformly across scales. The vortex-stretching term $(\omega \cdot \nabla) \mathbf{P}$ remains untouched: the fixed-regime model does not remove the central three-dimensional difficulty of the classical equations, and no regularity claim is made. Any smoothing must come from ν, λ , not from the conservative restoring channel — consistent with Results 1 and 2.

11 Pressure as absorption of the longitudinal current

The constraint $\nabla \cdot \mathbf{P} = 0$ makes $\mathbf{P}$ transverse. Any coarse current $\mathbf{J}$ built from the dynamics splits uniquely (Helmholtz, under the stated boundary conditions) as $\mathbf{J} = \mathbf{P} + \nabla \psi$ into a transverse part and a longitudinal part. Taking the divergence of the $\mathbf{P}$ -equation gives the pressure closure

$$\Delta \Pi = -\nabla \cdot [(\mathbf{P} \cdot \nabla) \mathbf{P}] + c^2 \Delta (\nabla \cdot \mathbf{D}) + \nabla \cdot \mathbf{S},$$

so the longitudinal component never appears as flow: it is absorbed instantaneously by the pressure. Equivalently $\mathbf{P} = \mathbb{P}_{\text{Leray}} \mathbf{J}$, the Leray projection applied to $\mathbf{J}$ (not to $\mathbf{P}$, which is already transverse). The decomposition holds in every regime; whether incompressibility itself can be derived from a deeper principle rather than imposed is left open.

12 Numerical study

Suppressing advection, pressure, and vector structure gives the scalar core

$$\partial_{tt} D - \nu \Delta \partial_t D + \lambda \partial_t D - c^2 \Delta D = S, \quad (4)$$

whose corners are exactly the rows of the closure table. When $\lambda > 0$, the formal overdamped limit $\lambda \rightarrow \infty$ with $\mu = c^2/\lambda$ fixed gives $\partial_t D \approx \mu \Delta D$: a diffusion law. In this limit only μ is observable; c^2 and λ are individually degenerate, and resolving them requires the inertial (wave) sector — precisely where the band of Section 8 lives. The band is thus both the sharpest prediction and the identifiability frontier.

Verification. The $k = 1$ mode on $[0, 2\pi]$, $D = q(t) \sin x$, obeys $\ddot{q} + (\lambda + \nu)\dot{q} + c^2 q = 0$. With a high-order integrator (relative tolerance 10^{-12}) and canonical parameters $\nu = 0.1$, $c = 1$, $\lambda = 0.4$ (so $\Lambda = 25$):

- conservation ($\nu = \lambda = 0$): relative energy drift $< 5 \times 10^{-12}$ over $t \in [0, 40]$;
- dissipation ($\nu = 0.1, \lambda = 0.4$): energy non-increasing to machine precision, final ratio 2.7×10^{-9} ;
- overdamped diffusion ($\lambda = 20, \mu = 0.05$): mode follows $e^{-\mu k^2 t}$ with relative error $< 2.6 \times 10^{-3}$ after the fast layer;
- band: $k_- = 0.2020$, $k_+ = 19.7980$; the $k = 1$ mode is in-band and rings at $\Omega_1 = \sqrt{1 - (0.25)^2} = 0.9682$ under $e^{-0.25t}$; the inversion (3) holds exactly, $k_- + k_+ = 20.0 = 2c/\nu$, $k_- k_+ = 4.0 = \lambda/\nu$;
- over-determination: Q -peak at $\sqrt{\lambda/\nu} = 2.0 = \sqrt{k_- k_+}$ with $Q_{\max} = 2.5 = \sqrt{\Lambda}/2$; frequency-peak at $\sqrt{2c^2 - \nu\lambda}/\nu = 14.0$ with $\Omega_{\max} = 9.798$;
- transfer bound: on a two-mode field, with X evaluated from its definition $c^2 \int \nabla \omega : \nabla \Omega_{\mathbf{D}}$, $|X| \leq c \|\nabla \omega\| \sqrt{2G}$ holds at all times, saturating when $\nabla \omega \parallel \nabla \Omega_{\mathbf{D}}$; a preloaded run ($G(0) > 0$, $\omega(0) = 0$) gives $X(0) = 0$ exactly and a release that grows only as $\nabla \omega$ develops.

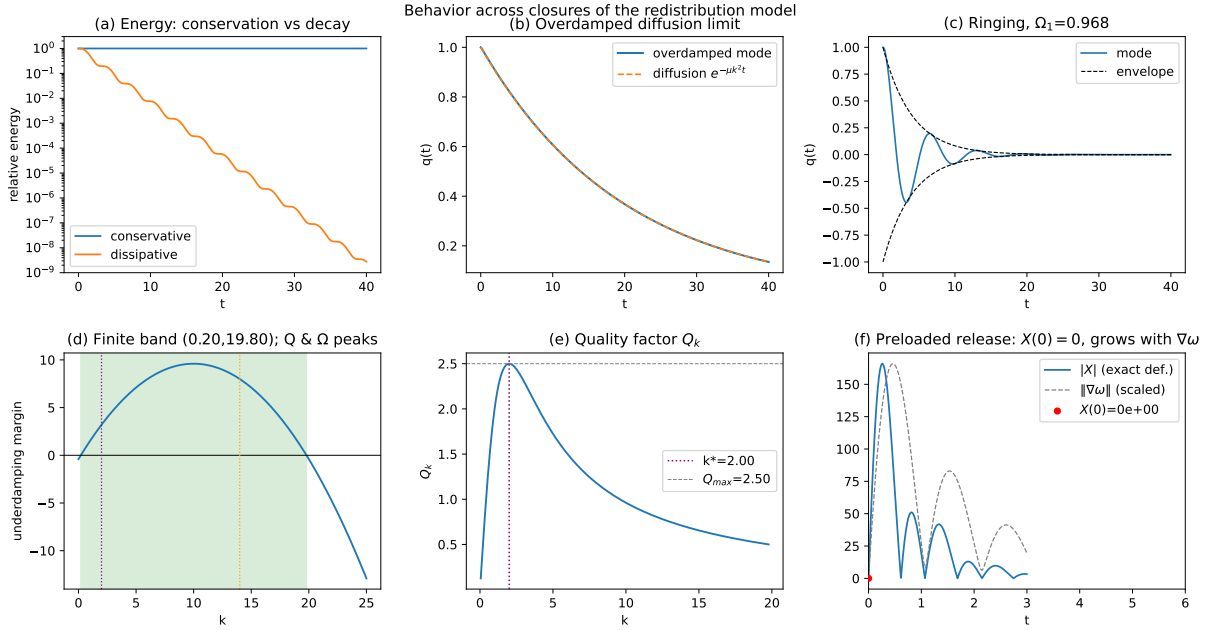


Figure 1: Behavior across closures. (a) conservative energy constant, dissipative energy monotone; (b) overdamped mode tracks the diffusion law; (c) underdamped ringing under its envelope; (d) the underdamping margin positive exactly on the finite band, with Q -peak (purple) and Ω -peak (orange) marked; (e) the quality factor Q_k peaking at $\sqrt{\lambda/\nu}$ with height $\sqrt{\Lambda}/2$; (f) a preloaded run ($G(0) > 0$, $\omega(0) = 0$): the transfer power starts at $X(0) = 0$ exactly and grows only as the active-vorticity structure $\|\nabla \omega\|$ develops, the soft release of stored orientation.

12.1 Two-dimensional incompressible vector validation

The scalar checks above are complemented by a full two-dimensional incompressible spectral solver for (1) on a periodic box, with Leray projection enforcing $\nabla \cdot \mathbf{P} = 0$ at every step (residual

$|\nabla \cdot \mathbf{P}| < 10^{-29}$), Taylor–Green initial velocity, and the accumulated field integrated from $\mathbf{D}_0 = 0$. Four statements are confirmed in genuinely vector, advective, inhomogeneous flow, where the vortex-stretching term is present and active.

Newtonian recovery. At $c = \lambda = 0$ the solver reproduces incompressible Navier–Stokes: the energy balance $dE/dt = -\nu\|\nabla\mathbf{P}\|^2$ holds to numerical tolerance.

Vector energy identity. With storage active ($c^2 = 0.5$, $\lambda = 0.1$, $\nu = 0.02$), the full energy $E = \frac{1}{2}\int(|\mathbf{P}|^2 + c^2|\nabla\mathbf{D}|^2)$ decays through dissipation only, $dE/dt = -\nu\|\nabla\mathbf{P}\|^2 - \lambda\|\mathbf{P}\|^2$, confirming that $c^2\Delta\mathbf{D}$ is conservative in two-dimensional incompressible flow and not merely in the scalar mode. The residual is purely temporal: halving Δt halves the error ($7.9 \rightarrow 3.9 \rightarrow 1.9 \times 10^{-3}$ for $\Delta t = 10^{-3}, 5 \times 10^{-4}, 2.5 \times 10^{-4}$), first-order convergence consistent with the finite-difference estimate of dE/dt , so the continuum identity is exact.

Enstrophy identity. The combined enstrophy $W + G = \frac{1}{2}\|\boldsymbol{\omega}\|^2 + \frac{1}{2}c^2\|\nabla\boldsymbol{\Omega}_\mathbf{D}\|^2$ is monotone non-increasing in the unforced run, decreasing at every sampled step (maximum increment -1.6×10^{-7}), confirming Result 2 with the orientation store G conservative and ν, λ the only sinks, in the presence of vortex stretching.

Conformation energy law in inhomogeneous flow. The finite-deformation identity of Result 5 is confirmed beyond homogeneous shear: evolving the conformation tensor $\mathbf{C}$ by the upper-convected derivative in the spatially varying Taylor–Green velocity gradient, the conservative-limit identity $dW/dt = \int \boldsymbol{\tau} : \nabla\mathbf{P}$ holds with relative error 1.9×10^{-3} (finite-difference limited), while $\det\mathbf{C}$ is preserved to 10^{-16} . The exact energy bookkeeping of the conformation closure therefore holds in inhomogeneous flow, not only in the homogeneous test.

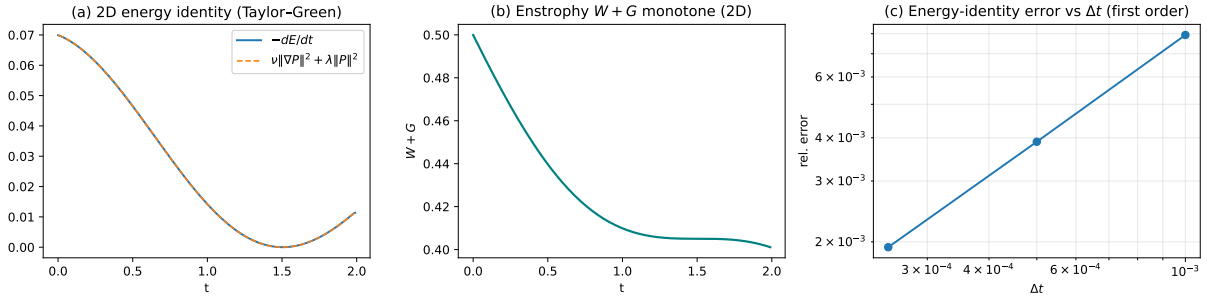


Figure 2: Two-dimensional incompressible validation. (a) the dissipation rate $\nu\|\nabla\mathbf{P}\|^2 + \lambda\|\mathbf{P}\|^2$ matches $-dE/dt$ in Taylor–Green flow with storage active; (b) the enstrophy $W + G$ is monotone non-increasing; (c) the energy-identity error is first order in Δt , confirming the continuum identity is exact and the residual numerical.

This closes the vector-validation, enstrophy-in-simulation, and inhomogeneous-conformation items of the gap register. The three-dimensional simulation, where vortex stretching can amplify rather than merely transport, remains the natural next stage.

13 Finite-deformation extension: the boundary of the small-deformation regime

Every result so far holds under the local rate law $\partial_t\mathbf{D} = \mathbf{P}$, declared in Section 2 as a small-to-moderate-deformation closure. That declaration marks a boundary internal to the model: the one hypothesis held provisional. This section crosses it. The model itself dictates what lives on the other side — not an external framework imposed on it, but the consequence of relaxing its own provisional law. The verdict has two parts: the obvious upgrade is impossible, and a specific alternative preserves an exact energy law at a stated cost.

13.1 The naive convected law is impossible

The natural finite-deformation upgrade keeps $\mathbf{D}$ a vector and makes its transport objective by an upper-convected derivative, $\partial_t \mathbf{D} + (\mathbf{P} \cdot \nabla) \mathbf{D} = \mathbf{P}$. In the one-dimensional loading reduction this is $D_t = P - P D_x$.

Result 4 (Energy defect of the naive convected law). *With the energy $E = \frac{1}{2} \int (|\mathbf{P}|^2 + c^2 |\nabla \mathbf{D}|^2)$ retained, the upper-convected vector law produces an extra energy-production term, absent under the local law, equal in the one-dimensional reduction to*

$$\left(\frac{dE}{dt} \right)_{extra} = -\frac{c^2}{2} \int_{\Omega} (\partial_x P) (\partial_x D)^2 dx,$$

generalizing in the incompressible vector case to $-\frac{c^2}{2} \int \nabla \mathbf{P} : (\nabla \mathbf{D} \otimes \nabla \mathbf{D})$. The term is cubic, sign-indefinite, and not a total time derivative; the exact energy identity fails and no conservative reinterpretation of the elastic store recovers it.

Derivation. Only the elastic store $\frac{1}{2} c^2 \|\nabla \mathbf{D}\|^2$ can change. Its density derivative is $c^2 \partial_x D \partial_x D_t$; under $D_t = P$ this is $c^2 \partial_x D \partial_x P$, the term cancelling against $c^2 \int P \Delta D$ in Appendix A. Under $D_t = P - P D_x$, $\partial_x D_t = P_x - P_x D_x - P D_{xx}$, adding $-c^2 P D_x D_{xx} - c^2 (D_x)^2 P_x$. Writing $-c^2 P D_x D_{xx} = -\frac{c^2}{2} P \partial_x [(D_x)^2]$ and integrating by parts (periodic) gives $+\frac{c^2}{2} \int P_x (D_x)^2$, so the net is $-\frac{c^2}{2} \int (\partial_x P) (\partial_x D)^2$. The sign of $\partial_x P$ is free, so the term injects or removes energy; D and $P = \partial_t D$ enter at different orders, so it has no antiderivative in t . $\square$

A spectral integration confirms the failure is not a small drift: with an undamped restoring core the energy diverges (relative excursion $\sim 10^5$ over a few time units, Figure 3, left). The naive objective upgrade is rejected.

13.2 A conformation tensor closes the budget

The defect is the signature of transporting the wrong object. Finite-deformation viscoelasticity transports not a displacement vector but a symmetric positive-definite *conformation tensor* $\mathbf{C}(\mathbf{x}, t)$ encoding the local strain state, by the upper-convected (Oldroyd) derivative

$$\overset{\nabla}{\mathbf{C}} := \partial_t \mathbf{C} + (\mathbf{P} \cdot \nabla) \mathbf{C} - (\nabla \mathbf{P}) \mathbf{C} - \mathbf{C} (\nabla \mathbf{P})^\top = -\frac{1}{\tau} (\mathbf{C} - \mathbf{I}) + (\text{source}),$$

with neo-Hookean (Oldroyd-B) free energy and elastic stress

$$W[\mathbf{C}] = \frac{c^2}{2} \int_{\Omega} (\text{tr } \mathbf{C} - \ln \det \mathbf{C} - d) d\mathbf{x}, \quad \boldsymbol{\tau} = c^2 (\mathbf{C} - \mathbf{I}).$$

The accumulated field of the main model is the small-deformation limit: with $\mathbf{C} = \mathbf{I} + 2\boldsymbol{\varepsilon} + O(\varepsilon^2)$ and $\boldsymbol{\varepsilon}$ the linear strain, $W \rightarrow \frac{c^2}{2} \int |\boldsymbol{\varepsilon}|^2$, reproducing $\frac{1}{2} c^2 \|\nabla \mathbf{D}\|^2$ up to the incompressible trace. The momentum balance carries $\nabla \cdot \boldsymbol{\tau}$ in place of $c^2 \Delta \mathbf{D}$.

Result 5 (Exact bookkeeping of the conformation law). *In the conservative limit ($1/\tau = 0$), the upper-convected conformation law satisfies*

$$\frac{dW}{dt} = \int_{\Omega} \boldsymbol{\tau} : \nabla \mathbf{P} d\mathbf{x} :$$

the rate of stored elastic energy equals exactly the mechanical power delivered by the flow through the elastic stress. With the kinetic balance this is a closed mechanical-energy identity in which W is a genuine conservative store and ν, λ are the only sinks.

Verification. For homogeneous shear of rate g , upper-convected transport reduces to $\dot{C}_{xx} = 2gC_{xy}$, $\dot{C}_{xy} = gC_{yy}$, $\dot{C}_{yy} = 0$. Numerically (Figure 3, right) $\det \mathbf{C}$ is conserved, $|\det \mathbf{C} - 1| < 10^{-14}$ (volume preservation, as incompressibility requires), and the stored-energy rate matches the input power, $\max |dW/dt - \tau_{xy}g| \approx 4 \times 10^{-4}$, at finite-difference precision. This is the finite-deformation analogue of the energy identity of Section 5, now valid at large strain.

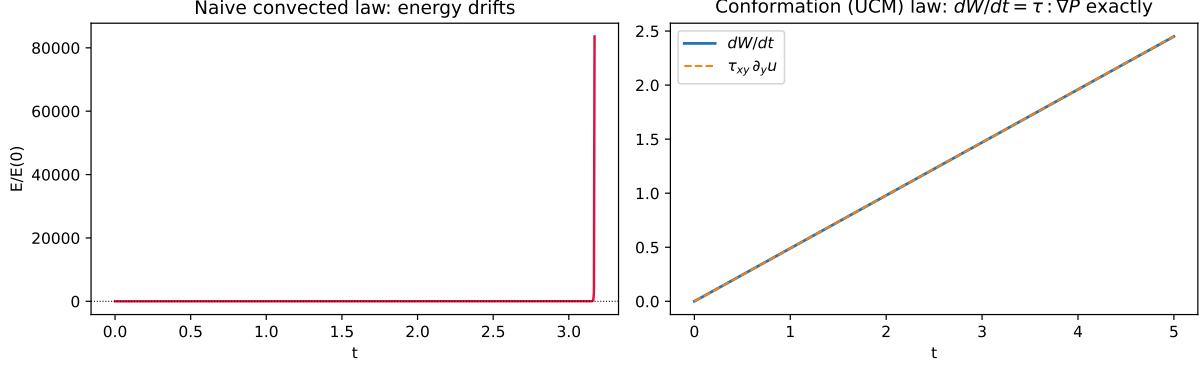


Figure 3: Left: the naive upper-convected *vector* law with the retained energy $\frac{1}{2}c^2\|\nabla \mathbf{D}\|^2$ does not conserve energy (it diverges; note the scale). Right: the conformation-tensor law conserves exactly, $dW/dt = \tau_{xy} \partial_y u$ to numerical precision, with $\det \mathbf{C}$ preserved.

13.3 Thermodynamic consistency with all dissipative channels

The conservative energy law of Result 5 covers the non-dissipative limit. The physically complete system adds Oldroyd-B relaxation, $\overset{\nabla}{\mathbf{C}} = -\frac{1}{\tau_r}(\mathbf{C} - \mathbf{I})$ with $\tau_r > 0$, alongside the two flow-dissipation channels ν, λ . The question is whether the coupled system remains thermodynamically consistent: whether a total free energy decreases monotonically in the unforced case, with nonnegative dissipation.

Result 6 (Free-energy dissipation of the coupled system). *Let $A(t) = \frac{1}{2}\int_{\Omega} |\mathbf{P}|^2 d\mathbf{x} + W[\mathbf{C}]$ with $W[\mathbf{C}] = \frac{c^2}{2}\int_{\Omega} (\text{tr } \mathbf{C} - \ln \det \mathbf{C} - d) d\mathbf{x}$. For smooth solutions of the coupled $\mathbf{P}$ - $\mathbf{C}$ system with all three dissipative channels active, under periodic or compatible no-flux boundary conditions,*

$$\frac{dA}{dt} = -\nu\|\nabla \mathbf{P}\|^2 - \lambda\|\mathbf{P}\|^2 - \frac{c^2}{2\tau_r} \int_{\Omega} \text{tr}[(\mathbf{C} - \mathbf{I})^2 \mathbf{C}^{-1}] d\mathbf{x} + \int_{\Omega} \mathbf{P} \cdot \mathbf{S} d\mathbf{x}.$$

All three dissipative terms are nonnegative; in particular the relaxation term satisfies, in the eigenbasis of $\mathbf{C}$ (eigenvalues $\lambda_i > 0$),

$$\text{tr}[(\mathbf{C} - \mathbf{I})^2 \mathbf{C}^{-1}] = \sum_i \frac{(\lambda_i - 1)^2}{\lambda_i} \geq 0,$$

vanishing if and only if $\mathbf{C} = \mathbf{I}$. Hence in the unforced case $dA/dt \leq 0$, with equality if and only if $\nabla \mathbf{P} = 0$, $\mathbf{P} = 0$, and $\mathbf{C} = \mathbf{I}$ simultaneously — thermodynamic equilibrium. The system is dissipative and free-energy-decreasing.

Derivation. The kinetic balance gives $\frac{d}{dt} \frac{1}{2}\|\mathbf{P}\|^2 = -\nu\|\nabla \mathbf{P}\|^2 - \lambda\|\mathbf{P}\|^2 + \int \mathbf{P} \cdot \nabla \cdot \boldsymbol{\tau} + \int \mathbf{P} \cdot \mathbf{S}$, the pressure term vanishing by incompressibility. For the elastic free energy, $\partial W/\partial \mathbf{C} = \frac{c^2}{2}(\mathbf{I} - \mathbf{C}^{-1})$ and $\boldsymbol{\tau} = c^2(\mathbf{C} - \mathbf{I})$; the transport-plus-stretching part of dW/dt equals $-\int \mathbf{P} \cdot \nabla \cdot \boldsymbol{\tau}$ exactly (the standard conformation stress-power identity, Result 5), cancelling the elastic power in the kinetic balance. The relaxation part contributes $\int \frac{c^2}{2}(\mathbf{I} - \mathbf{C}^{-1}) : (-\frac{1}{\tau_r}(\mathbf{C} - \mathbf{I})) = -\frac{c^2}{2\tau_r} \int \text{tr}[(\mathbf{C} - \mathbf{I})^2 \mathbf{C}^{-1}]$,

whose integrand is $\sum_i (\lambda_i - 1)^2 / \lambda_i \geq 0$ by the eigenvalue computation. Collecting terms gives the stated balance. $\square$

The result was verified numerically in homogeneous shear with relaxation active (Figure 4): the free-energy rate matches the input power minus the relaxation dissipation, $\max |dF/dt - (\text{power} - D_{\text{relax}})| \approx 9 \times 10^{-4}$, and $D_{\text{relax}} \geq 0$ holds at every step, reaching zero only as the system approaches its steady shear state. The coupled finite-deformation model is therefore thermodynamically consistent, closing the free-energy/entropy-production item of the gap register.

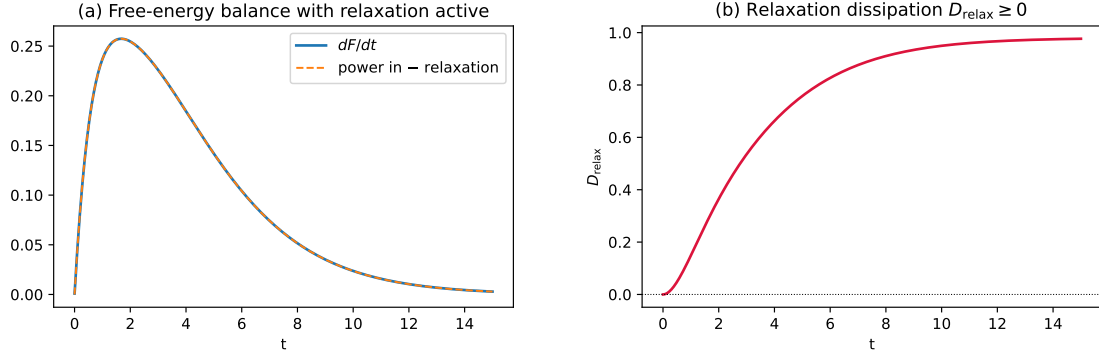


Figure 4: Thermodynamic consistency of the coupled **P**–**C** system with relaxation active. (a) the free-energy rate dF/dt equals the mechanical power in minus the relaxation dissipation; (b) the relaxation dissipation $D_{\text{relax}} = \frac{c^2}{2\tau_r} \text{tr}[(\mathbf{C} - \mathbf{I})^2 \mathbf{C}^{-1}]$ is nonnegative throughout, vanishing only at equilibrium.

13.4 What the extension reveals, and at what cost

Crossing the boundary declared in Section 2 returns a precise verdict: the naive route is impossible, breaking the budget irreparably; the conformation route succeeds, giving a finite-deformation form with an exact conservative energy law. The cost is intrinsic and is a change of primitive the model reveals about itself. In the small-deformation theory the accumulated field **D** is a displacement and **P** = $\partial_t \mathbf{D}$ its rate; at finite deformation the stored object is the conformation tensor **C**, a local strain state, of which the displacement is the small-deformation shadow. The interpretive reading “**D** is the displacement” (Section 16) holds only in the linear regime. Every linear result stands unchanged; the extension neither strengthens nor weakens them, and adds no claim about three-dimensional regularity, which the conformation formulation does not resolve. The free-energy/entropy-production consistency of the coupled system is established in Section 13.3, and the closed energy law is confirmed in inhomogeneous two-dimensional flow in Section 12.1. What remains genuinely open is the three-dimensional case, where vortex stretching can amplify rather than merely transport; it is now well-posed.

14 Falsifiers

The checks below are separated by level. Internal checks test whether the equations, code, and stated identities are mutually consistent. Empirical checks test whether a real medium belongs to this continuum class.

Internal falsifiers

The mathematical formulation or its implementation must be revised if:

1. it fails to return Navier–Stokes at $c = \lambda = 0$;
2. the energy identity (Result 1) fails in smooth unforced simulation under the stated boundary conditions;
3. the scalar overdamped diffusion limit cannot be recovered numerically;
4. the 2D enstrophy combination $W + G$ is not monotone non-increasing in unforced simulation (Result 2);
5. the transfer bound $|X| \leq c \|\nabla \boldsymbol{\omega}\| \sqrt{2G}$ is exceeded (Proposition 1);
6. with an empty orientation store ($\boldsymbol{\Omega}_{\mathbf{D}} \equiv 0$), the model releases a directed stored-orientation contribution immediately, contradicting $G = 0 \Rightarrow X = 0$;
7. with a preloaded orientation store $G(0) > 0$ but initially structureless active vorticity $\boldsymbol{\omega}(0) = 0$, the simulation produces a nonzero instantaneous transfer $X(0)$ or a jump-like release before $\nabla \boldsymbol{\omega}$ develops, contradicting Proposition 1. The transfer power must be evaluated from its definition $X = c^2 \int \nabla \boldsymbol{\omega} : \nabla \boldsymbol{\Omega}_{\mathbf{D}}$; estimating it as a finite difference of $G(t)$ introduces an endpoint bias that mimics a nonzero $X(0)$ where the exact value is zero, and is not a genuine violation;
8. in the finite-deformation extension (Section 13), the conformation transport fails to preserve $\det \mathbf{C}$ in incompressible flow, or the conservative-limit identity $dW/dt = \boldsymbol{\tau} : \nabla \mathbf{P}$ fails in smooth simulation.

Empirical falsifiers

A calibrated physical medium should not be described by the full two-channel storage closure if:

1. a medium independently shown to have both dissipative channels and an elastic storage channel never exhibits the two-cutoff band when $c^2 > \nu\lambda$;
2. measured band edges $k_{\pm}$ and in-band frequencies Ω_k return mutually inconsistent (ν, c, λ) ;
3. the observed band has only one cutoff when the inferred parameters require two, or two cutoffs when a one-channel limit is independently established;
4. the envelope decay rate inside the band is not affine in k^2 with intercept $\lambda/2$ and slope $\nu/2$;
5. the over-determination web (Section 8.2) fails: two independent measurements return incompatible parameters beyond experimental uncertainty.

15 Gap register

Item	Status	Comment
Objective finite-deformation transport of $\mathbf{D}$	resolved, with named cost	naive convected law impossible (Result 4); conformation-tensor law closes the energy budget (Result 5) at a change of primitive ($\mathbf{D} \rightarrow \mathbf{C}$)
Thermodynamic consistency of $\mathbf{P}$ – $\mathbf{C}$ with all dissipative channels	resolved	free-energy A monotone non-increasing; relaxation dissipation $\propto \sum_i (\lambda_i - 1)^2 / \lambda_i \geq 0$, zero only at $\mathbf{C} = \mathbf{I}$ (Result 6, §13.3)
Closed energy law in inhomogeneous flow	resolved (2D)	conformation identity $dW/dt = \int \boldsymbol{\tau} : \nabla \mathbf{P}$ confirmed in inhomogeneous Taylor–Green, $\det \mathbf{C}$ preserved to 10^{-16} (§12.1); 3D open
Boundary-condition catalogue	closed for periodic; open for general boundaries	theorems are stated for periodic or explicitly compatible boundaries; a full boundary taxonomy remains optional
Transverse accumulated field	closed by assumption	impose $\nabla \cdot \mathbf{D}_0 = 0$ for the studied sector; longitudinal preload is a decoupled elastic-energy channel, not part of the vorticity/band analysis
Vector incompressible validation (2D)	resolved	2D spectral solver confirms Newtonian recovery, the vector energy identity (first-order Δt convergence), and incompressibility ($ \nabla \cdot \mathbf{P} < 10^{-29}$); see §12.1
Three-dimensional validation	open	3D run, where stretching can amplify, not yet performed
Finite band against data	stated, untested	closed-form band and inversion given; not yet measured
Enstrophy identity in 2D simulation	resolved	$W + G$ monotone non-increasing in Taylor–Green flow (§12.1)
Energy/Helmholtz split	definitional	decomposition, not an independent result

16 An optional interpretation (detachable)

Nothing in Sections 1–15 depends on this section; it may be removed without affecting any definition, result, or proof. It records one reading that the established mathematics admits.

Under the rate law $\partial_t \mathbf{D} = \mathbf{P}$, the accumulated field is an initial stored configuration plus the time integral of the flow at a fixed point — in the small-deformation regime, the displacement field (at

finite deformation, by Section 13, the displacement is only the linear shadow of a conformation-tensor strain state). The model is then, in that regime, an incompressible medium with one elastic channel (stiffness c^2) and two possible dissipative channels (ν on rate gradients, λ on rate). The transverse restriction used in the main text selects the incompressible shear/orientation sector; a longitudinal preload would represent an additional compressive elastic store that is invisible to vorticity and not treated here. If $c = 0$ but $\lambda > 0$, the interpretation changes: the model is only a drag-modified incompressible flow, not a storage-and-return medium. This locates the linear storage sector inside a physically realized class and makes every prediction in principle measurable there.

Read mechanistically, the active field is the rate at which an accumulated quantity is redistributed under incompressibility, and the pressure is the closure that absorbs whatever part of a relational current is not transverse. In this reading the restoring channel is the restoration of accumulated difference, the friction is its bulk relaxation, and the enstrophy store $G = \frac{1}{2}c^2\|\nabla\mathbf{\Omega}_D\|^2$ is accumulated orientation held in memory: a vortex referencing its own axis draws on this store, a collimated release expresses it, and Proposition 1 caps the rate of either, forbidding release from an empty store. These are reorientations within a fixed space — selections of a direction, not generation of a new axis — and the word “redistribution” should be read as transfer between the stored and active channels, never as creation, since the balances of Sections 5–6 forbid it. The reading commits to nothing about what the accumulated quantity ultimately is; the model is built so that this can stay open while the mathematics stands closed. A separate companion manuscript in this program explores one candidate ontology for that open question — a relational-radius origin for the wave-speed c , from which an octave structure and a distinct falsifiable signature are derived; it is a deliberately more speculative document, and the present paper stands independently of it. The quantitative point of contact is that the dimensionless control number $\Lambda = c^2/(\nu\lambda)$ of the band here counts the octave thresholds there. Readers interested only in the continuum model need not consult it; readers interested in a possible deeper origin will find it as the geometric companion.

A Energy identity

Pairing the $\mathbf{P}$ -equation in (1) with $\mathbf{P}$ in L^2 : by incompressibility $-\int \mathbf{P}\cdot\nabla\Pi = \int \Pi\nabla\cdot\mathbf{P} = 0$ and $\int \mathbf{P}\cdot(\mathbf{P}\cdot\nabla)\mathbf{P} = \frac{1}{2}\int(\mathbf{P}\cdot\nabla)|\mathbf{P}|^2 = 0$; integration by parts gives $\nu\int \mathbf{P}\cdot\Delta\mathbf{P} = -\nu\|\nabla\mathbf{P}\|^2$ and, using $\mathbf{P} = \partial_t\mathbf{D}$,

$$c^2\int_{\Omega}\mathbf{P}\cdot\Delta\mathbf{D} = -c^2\int_{\Omega}\nabla\mathbf{P}:\nabla\mathbf{D} = -\frac{c^2}{2}\frac{d}{dt}\int_{\Omega}|\nabla\mathbf{D}|^2.$$

Collecting terms gives Result 1; the restoring contribution is a perfect time derivative, never a sink.

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